

COURSE TITLE : WOOD PRODUCTS-I
COURSE CODE : 4123
COURSE CATEGORY : B
PERIODS PER WEEK : 4
SEMESTER : 4
PERIODS PER SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
I	History of Plywood Industry and its role in Wood Industry	3
	Peeling and slicing of Veneer	4
	Processing of Veneer	6
	Test	1
II	Jointing and matching of Veneer	3
	Gluing and Pressing	5
	Dimensioning and sanding	5
	Test	1
III	Grading of Plywood	4
	Special Purpose plywood	4
	Properties and Testing	5
	Test	1
IV	Solid Bending	7
	Laminated wood Bending	6
	Test	1
Total		56

Course Outcome

Module	G.O	Student will be able to:
1	1	Understand Manufacture of Veneer In Plywood Industry
	2	Understand Processing of Veneer
2	1	Understand Jointing and Matching of Veneer
	2	Understand Gluing And Pressing of Veneer
	3	Understand Dimensioning and Finishing of Plywood
3	1	Comprehend Grading of Plywood
	2	Understand The Testing And Properties of Plywood
4	1	Understand Solid Wood Bending
	2	Understand Machine Bending
	3	Understand Laminated Wood Bending

SPECEFIC OUTCOME

MODULE-I

1.1.0 Understand Manufacture Of Veneer In Plywood Industry

- 1.1.1 Define the plywood, veneer, flitches, peeling, slicing and drying
- 1.1.2 Describe the advantage of plywood over solid wood
- 1.1.3 Explain the history of plywood manufacture
- 1.1.4 Describe the factors considering in the selection of logs and its storage and transport
- 1.1.5 State the end coating and end protection devices
- 1.1.6 Describe about softening treatment of logs prior to boiling and slicing
- 1.1.7 Explain the flitches and their preparation
- 1.1.8 Illustrate tangential and radial cutting
- 1.1.9 Explain the veneer slicing and methods
- 1.1.10 Illustrate vertical slicing machine
- 1.1.11 Explain the rotary peeling
- 1.1.12 Describe the nose bar, knife arrangement on peeling lathe
- 1.1.13 Illustrate schematic diagram of log with respect to knife and nose bar

1.2.0 Understand Processing of Veneer

- 1.2.1 Describe handling of veneers and clipping
- 1.2.2 Describe advantage of veneer drying
- 1.2.3 Explain Jet dryer, roller dryer, band dryer, breather and Kiln dryer
- 1.2.4 Explain above dryers with the help of neat sketches
- 1.2.5 Understand the processing of veneers
- 1.2.6 Explain storage and handling of veneers from peeling
- 1.2.7 Explain joining and splicing machine cross feed splicer

MODULE II

2.1.0 Understand Jointing and Matching of Veneer

- 2.1.1 State repair of veneers
- 2.1.2 Describe picus joint, scarf joint, lbus joint
- 2.1.3 Describe veneer matching
- 2.1.4 Illustrate , four piece matching, book matching side to side matching, end to end matching cathedral matching

2.2.0 Understand Gluing and Pressing of Veneer

- 2.2.1 Define adhesive, adhesion, adherent
- 2.2.2 Define glues
- 2.2.3 Describe glue mixing
- 2.2.4 Describe roller spreading with diagram, brushing, spraying.
- 2.2.5 State the factors affecting wood to wood glue joint such as temperature, moisture content.
- 2.2.6 Describe pre pressing and its advantages.
- 2.2.7 Explain cold press
- 2.2.8 Explain hot press

- 2.2.9 Estimate gauge pressure
- 2.2.10 State the need of conditioning
- 2.2.11 Describe hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps
- 2.2.12 Explain the dimensioning and sanding of plywood

2.3.0 Understand Dimensioning and Finishing of Plywood

- 2.2.13 State the sanding and dimensioning
- 2.2.14 Describe trimming
- 2.2.15 State standard plywood specifications
- 2.2.16 Describe drum sander, belt sander, wide belt sander
- 2.2.17 State abrasive materials

MODULE III

3.1.0 Comprehend Grading of Plywood

- 3.1.1 State the grading of general purpose plywood
- 3.1.2 Describe the grading with respect to quality office veneers
- 3.1.3 Describe the grading with respect to adhesive used

3.2.0 Understand the Testing and Properties of Plywood

- 3.2.1 State speciality plywood
- 3.2.2 List speciality plywood.
- 3.2.3 Describe the tea chest plywood, marine plywood shuttering plywood, fireproof plywood, aircraft plywood, preservative treated plywood metal faced and moulded plywood
- 3.2.4 State the grading of specialty plywood by using relevant is codes

MODULE IV

4.1.0 Understand Solid Wood Bending

- 4.1.1 Define Bending
- 4.1.2 Explain General Consideration in bending
- 4.1.3 Explain steaming
- 4.1.4 Explain cold bending and Hot Bending
- 4.1.5 Explain simple 'U' bend with figure
- 4.1.6 Explain Re entrant and 'S' type bend with figure
- 4.1.7 Explain end stops
- 4.1.8 Explain chair leg bending

4.2.0 Understand Machine Bending

- 4.2.1 State Machine Bending
- 4.2.2 Explain Machine bending

- 4.2.3 Explain Rope and windless Machine with figure
- 4.2.4 Explain Lever arm Machine with figure
- 4.2.5 Explain Revolving Table Machine
- 4.2.6 Explain setting technique
- 4.2.7 Explain Movement of Bends

4.3.0 Understand Laminated Wood Bending

- 4.3.1 Define laminated bending
- 4.3.2 Explain selection and Preparation of laminates
- 4.3.3 Define Pre bending
- 4.3.4 Explain laminated wood bending by male & female forms
- 4.3.5 Explain split female forms, by inflated flexible hose
- 4.3.6 Explain two plane bends

CONTENT DETAILS

MODULE-I

Introduction – uses and advantageous of plywood – history of plywood making – world production and analysis of consumption- Selection of logs – storage – pre – treatments of logs – boiling and steaming – flitches and their preparation – tangential and radial cutting of veneers – veneer slicing – vertical slicing and horizontal slicing. Rotary cutting – lathe parts – position of knife and nose bar on a peeling lathe with respect to log.

Objective of veneer drying – roller dryer, jet dryer, band dryer, breather dryer, kiln dryer conventional drying system – physics of veneer drying. Storage of veneers.

MODULE-II

Jointing machine, taping, splicing machine, end jointing, scarf joint, Ibus joint, repair of veneers. Matching of veneers – four piece matching, book matching, side to side matching, end to end matching, cathedral matching – veneer sorting

Define - adhesive, adhesion, adherent, classification of glues with their chemical composition glue mixing list of additives used in mixing, methods of glue spreading – roller spreading, brushing, spraying. Assembling of glued veneers Pre – pressing advantages and disadvantages factors which affect the adhesion on wood to wood glue joint - moisture content - glue spread – temperature - pressure – surface porosity. Cold pressing and hot pressing. Calculation of gauge pressure. Conditioning of plywood. Hot press gluing faults – blister – bleed through - - warp age – starved joint - spotty bond – weak bondface checking – overlaps and gaps Trimming sanding – drum sander – belt sander – wide belt sander –

MODULE-III

Grading of Plywoods Physical properties – density, moisture – content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties – elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code – determination of density, moisture content – glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Teacheest plywood, marine plywood, shutting plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood.

MODULE-IV

Solid Bending- General Considerations- Chemical treatment- Selection of Species- softening treatment- Treating Methods- Bending in Cold State- Hot bending- Simple ‘U’ Bending – Re entrant and ‘S’ type bend- chair leg bends- Transverse bending – Built up bends Machine bending- Rope and windless machine- Lever arm Machine- Revolving Table Machine- Presses- Setting techniques – Movement of bends Laminated Wood Bending Laminated bending- General Consideration- Advantages- Disadvantages- selection and preparation of laminates – Pre bending- Pressing laminates in to shape- General Considerations- Male & Female forms, Split female forms- pre bend- pressing laminations by shape by Inflated flexible hose – Continuous strip Method- Two plane bends— Curved laminated Members- Movement and distortion of laminated bends- Plywood bending

REFERENCE

1. F.P. Kollmann, Principles of wood science and technology Vol.I, II – Springer Verlag
2. W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office
- 3.. Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore.